**Question:**

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition?

Answer Choices:

A. It activates TLR5 on the surface of macrophages.

B. It primarily induces the Th1-cell response.

C. Peptidoglycan is its major antigen that induces an immune response.

D. This pathogen evades the immune response by encapsulation.

**Reasoning + Answer:**

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To determine the immune response elic ited by the path ogen causing the patient 's condition , we need to consider the characteristics of the path ogen and the symptoms presented . The patient has been diagnosed with community -ac quired pneumonia , which is often caused by bacteria such as Stre pt oc occus pneumonia e . This type of pneumonia typically involves a robust inflammatory response , and the symptoms suggest a bacterial infection . The path ogen in question likely triggers an immune response that involves the activation of various immune cells and the production of cytok ines . Considering the symptoms and the treatment with c eph alex in , a beta -l act am antibiotic , it is likely that the path ogen is a Gram -positive bacter ium , such as Stre pt oc occus pneumonia e . Gram -positive bacteria have a thick pe pt id og ly can cell wall , which is a major antigen that can induce an immune response . This suggests that the correct answer is related to the pe pt id og ly can antigen .</ think >< answer > C . Pe pt id og ly can is its major antigen that induces an immune response .